



Incident handler's journal

Date: 10/03/2026	Entry: 1: Ransomware attack
Description	Documenting a ransomware security incident at a small U.S. health care clinic. The attack encrypted the organization's computer files and severely disrupted business operations.
Tool(s) used	None
The 5 W's	• Who: An organized group of unethical hackers known to target health care and transportation organizations. • What: Ransomware was deployed on the clinic's computers, encrypting critical files, including medical records. The attackers left a ransom note demanding a large sum of money in exchange for the decryption key. • When: Tuesday at approximately 9:00 a.m. • Where: At a small health care clinic in the United States. • Why: The attackers gained access through targeted phishing emails sent to several employees. The emails contained a malicious attachment that installed malware when downloaded and their motive was to get paid for the decryption key.
Additional notes	• Should the clinic pay the ransom? Paying does not guarantee the files will be decrypted. • Does the clinic have backups to restore the files? • Because medical records were affected, the clinic may need to report the breach under HIPAA.

Date: 10/03/2026	Entry: 2: Analyzing a packet capture with Wireshark
Description	Analyzed a sample packet capture file using Wireshark to practice inspecting network traffic and applying filters to isolate relevant packets.
Tool(s) used	Wireshark, an open-source network protocol analyzer with a graphical user interface. Security analysts use Wireshark to capture and examine network traffic, which helps detect suspicious activity, troubleshoot network issues and investigate security incidents.
The 5 W's	• Who: N/A • What: N/A • When: N/A • Where: N/A • Why: N/A
Additional notes	• Opened a packet capture (.pcap) file and reviewed the packet list, packet details and packet bytes panes. • Applied display filters to narrow the traffic by IP address, protocol and port, including filtering for DNS and web traffic. • Inspected packet details such as source and destination IP addresses, ports and protocol information. • At first, the amount of data in Wireshark was overwhelming. Filters were key to finding relevant traffic quickly. • Next step: practice with real-world capture files to improve at spotting malicious traffic.

Date: 10/04/2026	Entry: 3: Capturing network traffic with tcpdump
Description	Captured and analyzed live network traffic using tcpdump in a Linux command-line environment, then saved the traffic to a packet capture file for later analysis.
Tool(s) used	tcpdump, a command-line network protocol analyzer. Like Wireshark, it lets analysts capture, filter and analyze network traffic. Because it runs from the command line, it's lightweight and useful on servers without a graphical interface.
The 5 W's	• Who: N/A • What: N/A • When: N/A • Where: N/A • Why: N/A
Additional notes	• Identified available network interfaces with <code>sudo ifconfig</code> and <code>sudo tcpdump -D</code>. • Captured live traffic on the <code>eth0</code> interface with <code>sudo tcpdump -i eth0 -v -c5</code>. • Filtered for web traffic on port 80 and saved it to a file (<code>capture.pcap</code>) using the <code>-w</code> option, then generated traffic with <code>curl</code>. • Read the saved capture with <code>-r</code>, and viewed packet contents in hex and ASCII with <code>-X</code>. • Learned that <code>-nn</code> turns off name and port resolution, which keeps the output cleaner and avoids extra DNS lookups during an investigation.

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10/04/2026	4: Investigating a suspicious file hash
Description	Investigated a suspicious file hash from a phishing alert to determine whether the file was malicious. This activity falls within the <i>Detection and Analysis</i> phase of the NIST incident response lifecycle.
Tool(s) used	VirusTotal, an investigative tool that analyzes files, URLs, domains and IP addresses against dozens of security vendors' detection engines. Analysts use it to quickly check whether an indicator of compromise (IOC) has been reported as malicious.
The 5 W's	• Who: An unknown malicious actor • What: An employee received an email with a password-protected spreadsheet attachment. After opening it with the password included in the email, a malicious payload executed on their computer. File hash (SHA-256): 54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b • When: At 1:20 p.m., the intrusion detection system (IDS) sent an alert to the SOC after detecting the file • Where: On an employee's computer at a financial services company • Why: The employee downloaded and opened a malicious attachment from a phishing email, which allowed the malware to run
Additional notes	VirusTotal confirmed the file hash as it was flagged by 54 security vendors and identified as malicious. The password-protected attachment was likely used to bypass email security scanning. Next steps: isolate the infected computer, escalate to a Level 2 analyst, and block the sender and file hash. Prevention: security awareness training on suspicious attachments, and email filtering that quarantines password-protected files from external senders

Reflections/Notes	Were there any specific activities that were challenging for you? Why or why not? • tcpdump was the most challenging because I'm new to the command line. Wireshark was overwhelming at first too, but I learned that filters make it easier to find what matters. Has your understanding of incident detection and response changed after taking this course? • Yes. I learned that incident response follows a step-by-step process, and that good documentation is key. My quality control experience has helped me see why accurate records matter. Was there a specific tool or concept that you enjoyed the most? Why? • VirusTotal, because it quickly showed whether a file was malicious. I want to keep practicing with these tools to build more confidence.
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